

$$\text{Form A} \quad \bar{a} = 1 - e^{-b(\bar{w} - \bar{w}_s)} \quad p = 0$$

$$\text{Form B} \quad \bar{a} = 1 - \frac{b}{(\bar{w} - \bar{w}_s) + b} \quad p = 1$$

$$\text{Form C} \quad \bar{a} = 1 - \frac{b}{1 + \bar{w} - \bar{w}_s} \quad p = 1$$

$$\bar{w}_s \text{ known,} \quad \bar{a}'_{\text{true}} := \frac{d\bar{a}_{\text{true}}}{d\bar{w}}$$

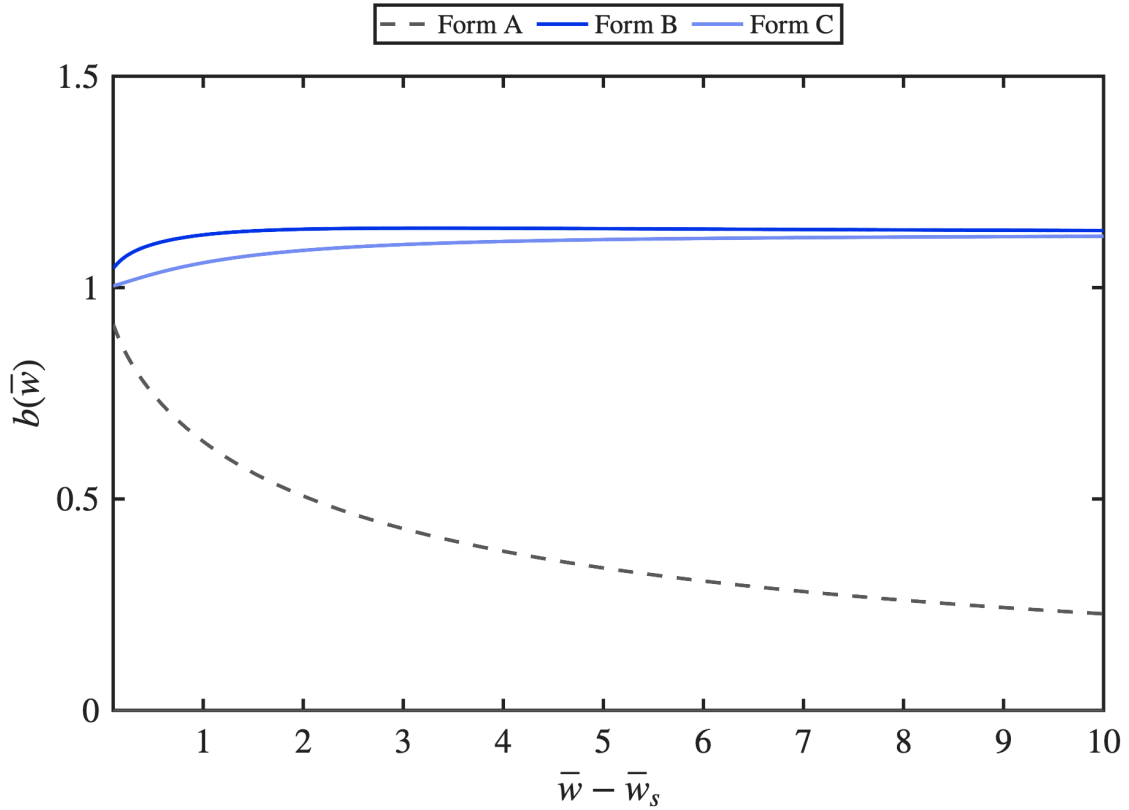
$$b_A(\bar{w}) = -\frac{\ln(1 - \bar{a}_{\text{true}})}{\bar{w} - \bar{w}_s}, \quad b_B(\bar{w}) = (\bar{w} - \bar{w}_s) \frac{1 - \bar{a}_{\text{true}}}{\bar{a}_{\text{true}}}, \quad b_C(\bar{w}) = (1 + \bar{w} - \bar{w}_s)(1 - \bar{a}_{\text{true}})$$

$$\bar{w} - \bar{w}_s \in [0.1, 10]$$

$$\bar{w} - \bar{w}_s = 0 : \quad \text{A, B : any } b ; \quad \text{C : } b = 1$$

$$\bar{w} - \bar{w}_s = \infty : \quad \text{A, B, C : any } b$$

$$\text{elsewhere : } \partial \bar{a} / \partial b \neq 0 \Rightarrow b(\bar{w}) \text{ unique}$$

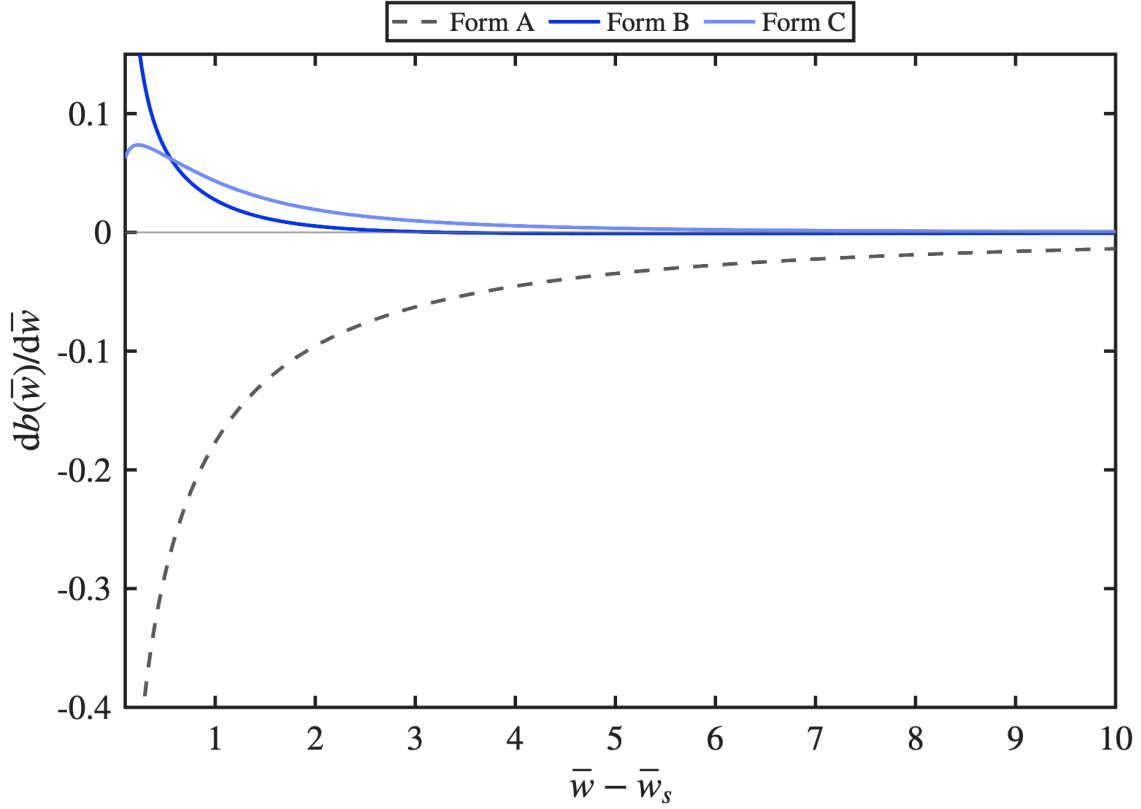


$$\frac{db_A(\bar{w})}{d\bar{w}} = \frac{1}{(\bar{w} - \bar{w}_s)^2} \left[\frac{(\bar{w} - \bar{w}_s) \bar{a}'_{\text{true}}}{1 - \bar{a}_{\text{true}}} + \ln(1 - \bar{a}_{\text{true}}) \right]$$

$$\frac{db_B(\bar{w})}{d\bar{w}} = \frac{1 - \bar{a}_{\text{true}}}{\bar{a}_{\text{true}}} - (\bar{w} - \bar{w}_s) \frac{\bar{a}'_{\text{true}}}{\bar{a}_{\text{true}}^2}$$

$$\frac{db_C(\bar{w})}{d\bar{w}} = (1 - \bar{a}_{\text{true}}) - (1 + \bar{w} - \bar{w}_s) \bar{a}'_{\text{true}}$$

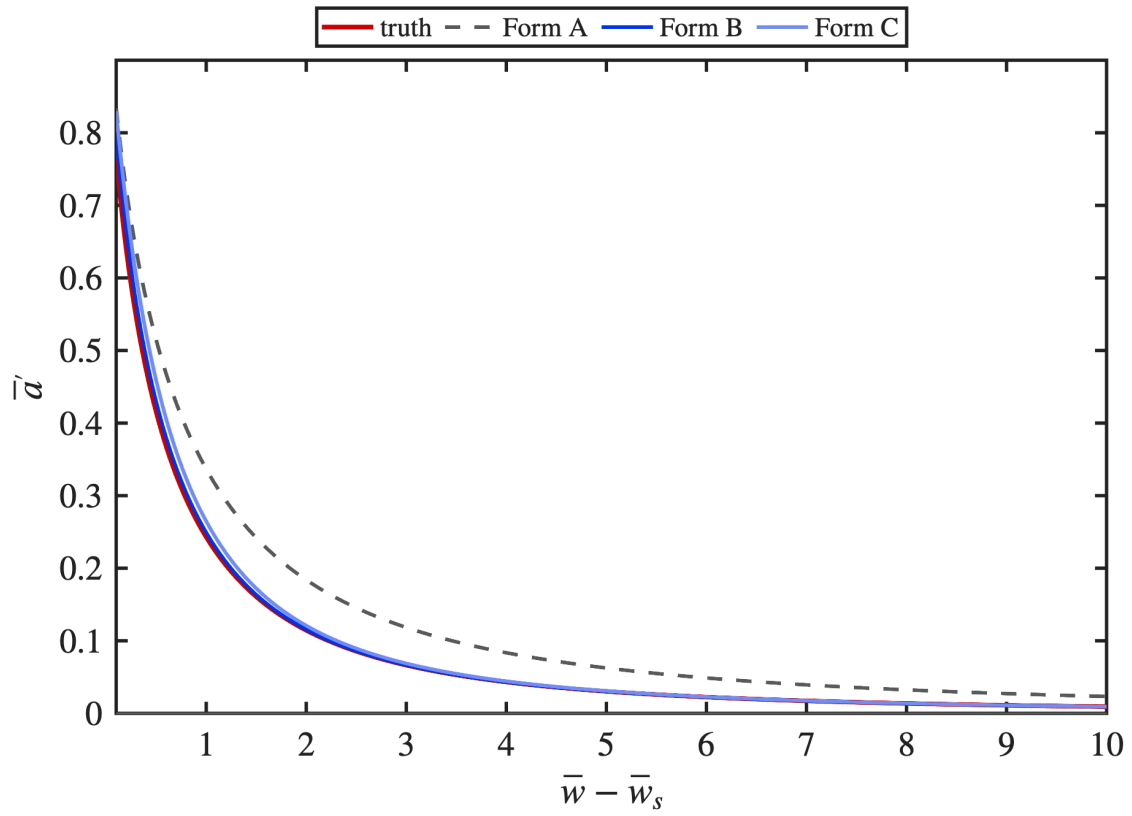
$$\left. \frac{db(\bar{w})}{d\bar{w}} \right|_{\bar{w} - \bar{w}_s = 0.1} = -0.669, +0.316, +0.063 \quad \left. \frac{db(\bar{w})}{d\bar{w}} \right|_{\bar{w} - \bar{w}_s = 10} = -0.014, -0.001, +0.001$$



$$\bar{a}'_A = -\frac{(1 - \bar{a}_{\text{true}}) \ln(1 - \bar{a}_{\text{true}})}{\bar{w} - \bar{w}_s}$$

$$\bar{a}'_B = \frac{\bar{a}_{\text{true}}(1 - \bar{a}_{\text{true}})}{\bar{w} - \bar{w}_s}$$

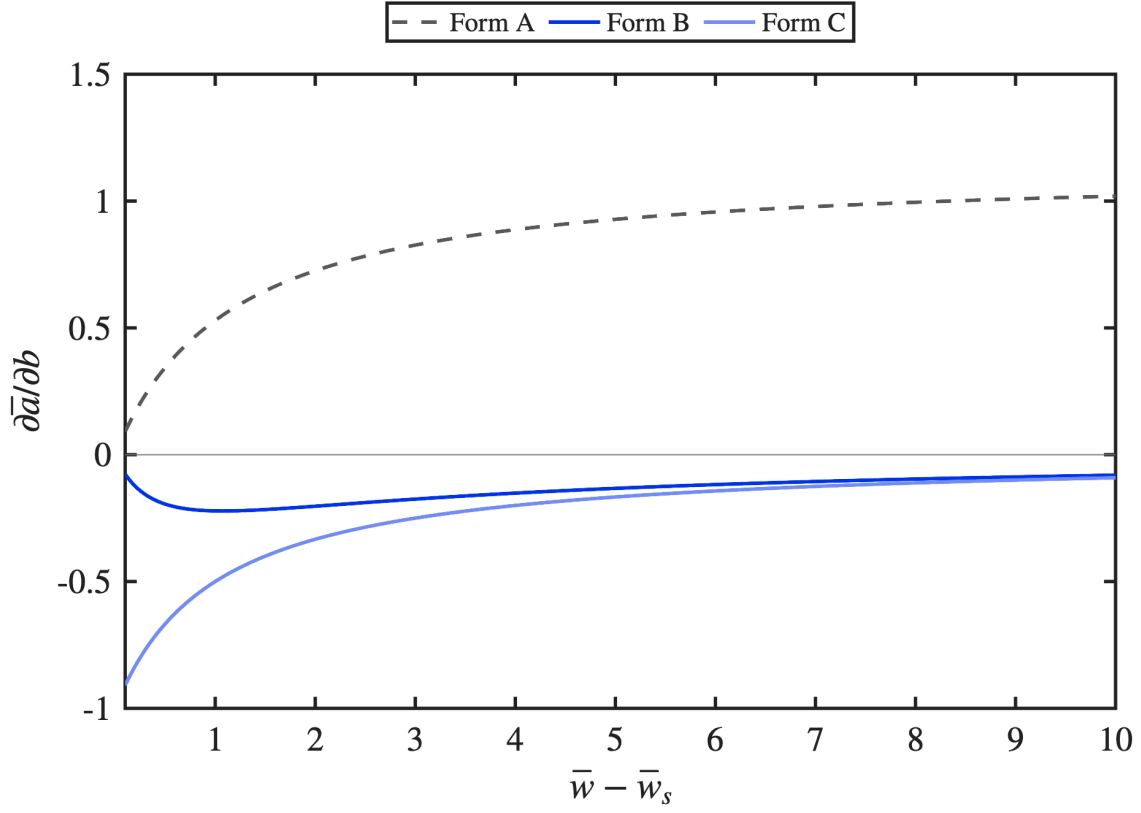
$$\bar{a}'_C = \frac{1 - \bar{a}_{\text{true}}}{1 + \bar{w} - \bar{w}_s}$$



$$\frac{\partial \bar{a}_A}{\partial b} = +(\bar{w} - \bar{w}_s)(1 - \bar{a}_{\text{true}})$$

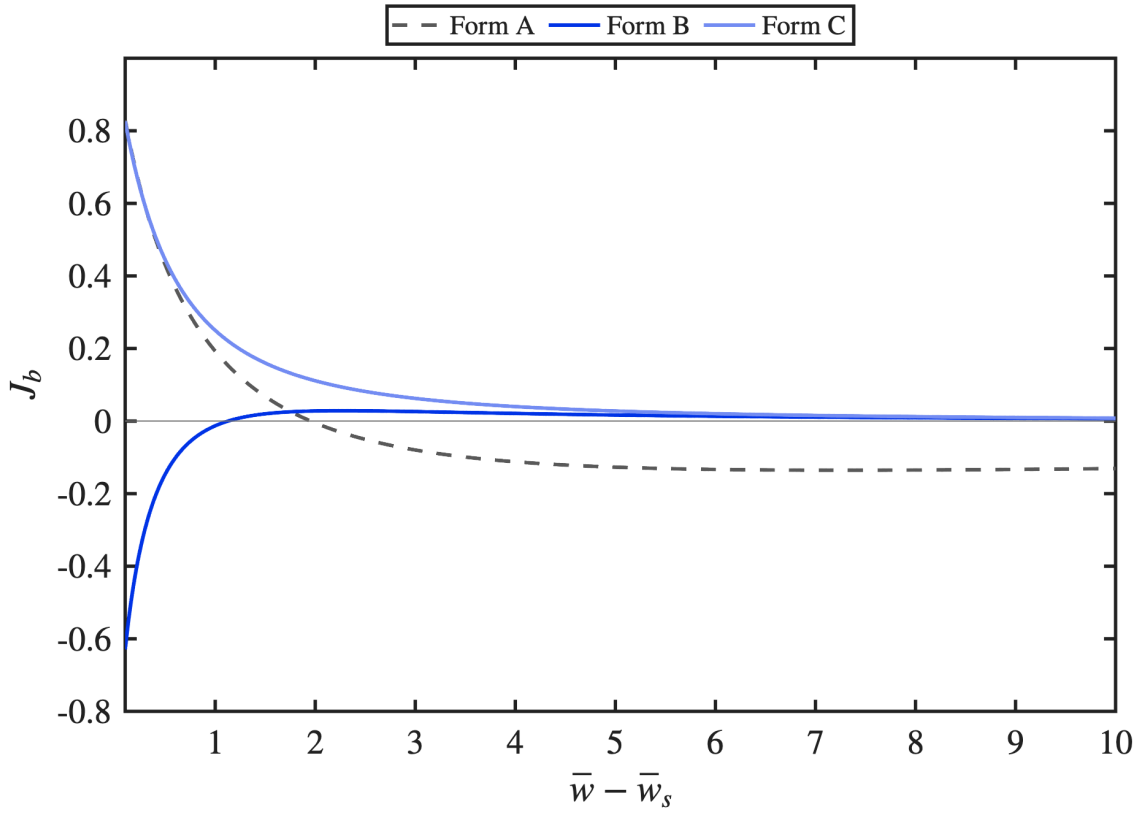
$$\frac{\partial \bar{a}_B}{\partial b} = -\frac{\bar{a}_{\text{true}}^2}{\bar{w} - \bar{w}_s}$$

$$\frac{\partial \bar{a}_C}{\partial b} = -\frac{1}{1 + \bar{w} - \bar{w}_s}$$



$$J_{b,A} = (1 - \bar{a}_{\text{true}}) \left[1 + \ln(1 - \bar{a}_{\text{true}}) \right] \quad J_{b,B} = \frac{\bar{a}_{\text{true}}^2 (2\bar{a}_{\text{true}} - 1)}{(\bar{w} - \bar{w}_s)^2} \quad J_{b,C} = \frac{1}{(1 + \bar{w} - \bar{w}_s)^2}$$

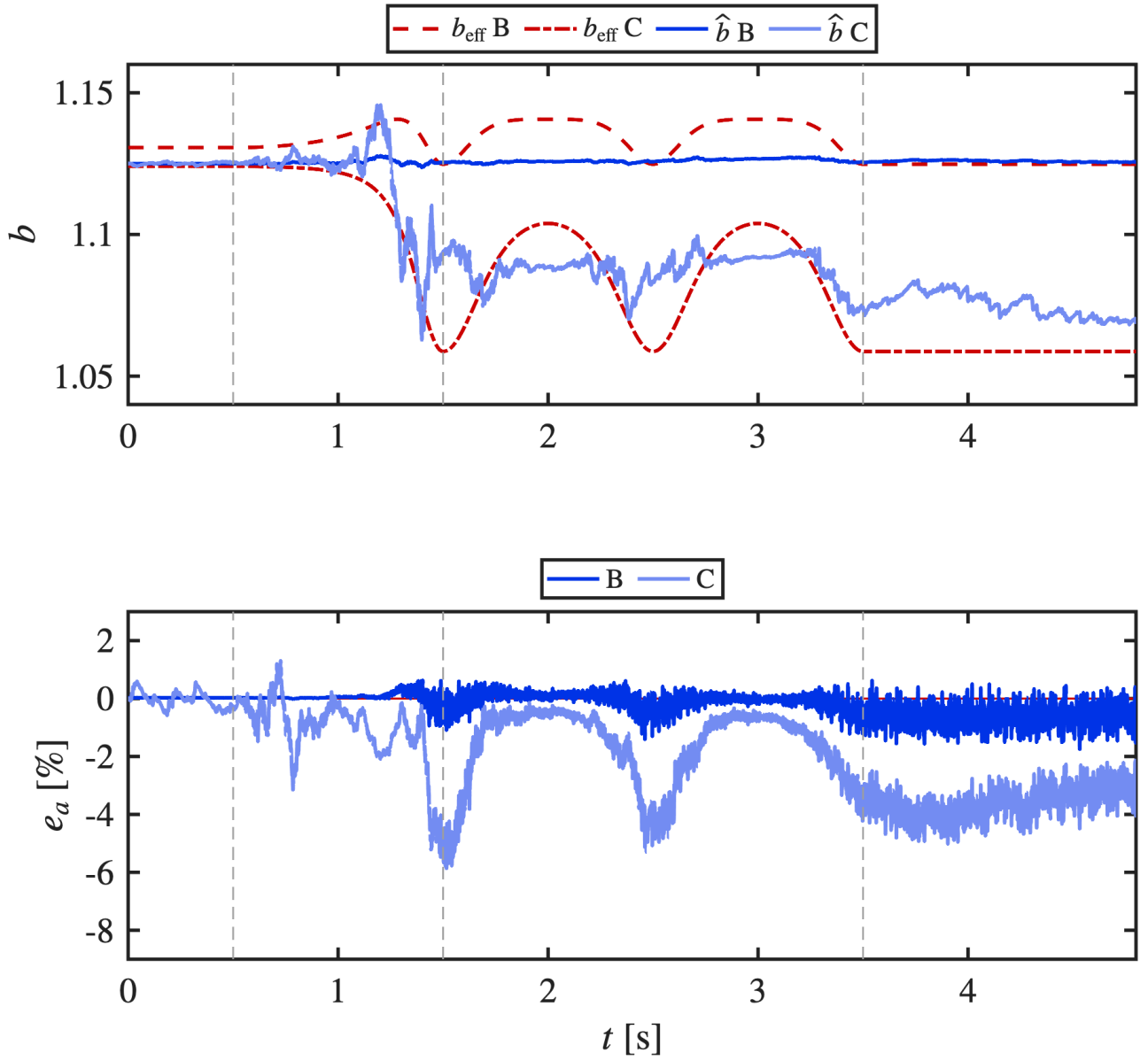
$$J_b = 0 : \quad \text{A : } \bar{a}_{\text{true}} = 1 - e^{-1}, \quad \bar{w} - \bar{w}_s = 1.955 \quad \text{B : } \bar{a}_{\text{true}} = \frac{1}{2}, \quad \bar{w} - \bar{w}_s = 1.128 \quad \text{C : none}$$



truth = $c_{\perp}$	$b_0 = \frac{9}{8}$ (both)	$p = 1, \bar{w}_s = 1$
$\hat{b}$ estimated	$\sqrt{P[0]} = 0.0157$ (B), 0.0708 (C)	one seed

$b_{\text{eff}} = \text{page 1}; \quad \sup_{\text{envelope}} |b_{\text{eff}} - \frac{9}{8}| = 0.0157$ (B), 0.0708 (C)

descent peak, osc RMS, final hold [%] B : 0.95, 0.26, -0.60 C : 5.47, 1.87, -3.52



truth = $c_{\parallel}$	$b_0 = \frac{9}{16}$ (both)	$p = 1, \bar{w}_s = 1$
$\hat{b}$ estimated	$\sqrt{P[0]} = 0.1880$ (B), 0.0072 (C)	one seed

$\sup_{\text{envelope}} |b_{\text{eff}} - \frac{9}{16}| = 0.1880$ (B), 0.0072 (C) swing = 0.1769, 0.0069

descent peak, osc RMS, final hold [%] B : 5.81, 2.00, -3.83 C : 0.91, 0.44, -0.58

$c_{\parallel}$ substituted into the wall-normal channel; $c_{\perp}$ has a pole at contact, $c_{\parallel}$ has none (3.084)

